

James Boutros

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EDUCATION

University of Waterloo

BASc in Electrical and Computer Engineering

Waterloo, ON

Sept. 2026 to Present

Assumption Catholic S.S.

OSSD: 98% Average, 4.0 Unweighted GPA

Burlington, ON

Graduated 2026

EXPERIENCE

Phydata | Co Founder and CTO

June 2026 to Present

5 Person Founding Team

Waterloo, ON

- Co founded physical AI data infrastructure for collection, labeling, and dataset management; secured \$20,000 in early investment and applied to Y Combinator W27.
- Designed the website, MVP roadmap, and scalable data collection hardware concepts in Autodesk Fusion 360.

University of Waterloo AI/ML Research

June 2026 to Present

Student Researcher

Toronto, CA (Remote)

- Study SHAP explanation drift under distribution shift across Logistic Regression, XGBoost, and TabPFN.
- Evaluated hospital readmission and U.S. Census housing data using AUROC, AUPRC, and calibration metrics; Preparing Submission for AAAI 2027.

UW ASIC | RTL Design and Verification

Sept. 2026 to Present

TinyTracer Team | Multiplier 2 RTL + Verification

Waterloo, ON

- Develop SystemVerilog RTL for TinyTracer's Multiplier 2, implementing pipelined fixed point arithmetic for a Tiny Tapeout ray tracing graphics chip.
- Verify signed products, rounding, saturation, and request/response interfaces against the arithmetic specification.

WATonomous | Electrical Engineer, Humanoid Robotics

Sept. 2026 to Present

Humanoid Robotics Team

Waterloo, ON

- Design separate CAN buses for 1 Mbps CubeMars motors and a 250 kbps BMS; evaluate harness length, topology, termination, signal degradation, EMI, and voltage drop.
- Develop a sensor stack board beside the power distribution board for IMU integration and state estimation.

iOS Application Developer

June 2025 to July 2025

Career Education Council / Apple Co op

Toronto, CA (Remote)

- Built MicroLoop in SwiftUI with micro habit tracking, AI goal breakdown, and streaks; presented to Apple engineers.

PROJECTS

AstronoJarvis + Daily Control Agent + Waterloo All in 1 | Python, TypeScript, Obsidian

- Built a Waterloo browser extension used by 100+ users, consolidating deadlines, schedules, email, and co op events.
- Customized my device's extension to relay information through Obsidian to AstronoJarvis and the Daily Control Agent, sharing context for briefs, task planning, and automation.
- Produced 120 research artifacts and surfaced 500+ co op roles with source auditing and duplicate safe updates.

Custom PCB Design Projects | Altium Designer, Schematic Capture

- Designed voltage regulator PCBs through two Altium onboarding projects, including an LM1117 adjustable 3.3 V supply; selected components and created custom symbols and footprints.
- Completed schematic capture, placement, routing, and DRC; considered thermal dissipation, capacitor requirements, power paths, connectors, and manufacturability.

Self Righting Robot | Soldering, ROS 2, MuJoCo

- Integrated robot hardware through mounting, wiring, soldering, sensor/actuator connections, and bring up.
- Contributed to ROS 2 and MuJoCo simulation, hardware interfaces, testing, and transfer to the physical robot.

TECHNICAL SKILLS

Languages: Python, C++, SystemVerilog, Swift, JavaScript/TypeScript, HTML/CSS

Hardware / Robotics: Altium, Fusion 360, CAN bus, IMUs, ROS 2, MuJoCo, Arduino/ESP32, soldering

Frameworks / Tools: SwiftUI, Next.js, React, Git, Vercel, Supabase, Obsidian, Playwright

Concepts: RTL verification, signal integrity, state estimation, embedded/distributed systems, REST APIs, model evaluation under distribution shift

ADDITIONAL

3x Hackathon Winner | Competitive Weightlifting: 3 Years | VEX Robotics Competitor: 3 Years